

Trisha Gumidelli

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EDUCATION

Mahindra University

B.Tech. Artificial Intelligence

Aug. 2026

Excellencia Junior College

Current GPA: 8.6/10.0

July 2022

COURSEWORK

Courses: Data Structures, Natural Language Processing, Digital Image Processing, Machine Learning & Python, Database Management Systems, Deep Neural Networks, Recommender Systems, Object-Oriented Programming in Java, Software Engineering.

SKILLS

Languages: C, Python, Java, HTML/CSS

Tools: Git/GitHub, VS Code, Jupyter Notebook, ChatGPT, Gemini, Google AI Studio, NotebookLM, n8n (workflow automation)

Productivity and Automation: Prompt Engineering, AI-assisted Development, Workflow Automation, AI-driven Data Analysis

Frameworks: React, Node.js, Flask, FastAPI

Libraries: pandas, NumPy, Matplotlib, Scikit-learn,

PROJECTS

Bio-Mind

Apr. 2025

- Developed an AI-driven protein/gene exploration platform using FastAPI, React, and biomedical APIs for NLP-based querying.
- Enabled structural, functional, and clinical comparisons of proteins, with OpenAI chatbot support for intelligent insights.
- Designed a responsive UI with React, TypeScript, and Bootstrap for enhanced user interaction.

Alternative Medicine Generator

Aug. 2024

- Developed a Flask-based web application that suggests alternative medicines using machine learning and NLP techniques.
- Utilized scikit-learn to vectorize medicine compositions with CountVectorizer and implemented a Nearest Neighbors model for similarity matching using cosine-similarity.
- Designed a user-friendly interface with HTML, CSS, and JavaScript, ensuring a responsive and interactive experience.

Human Pose Estimator for Fitness Applications

Aug 2024 – Dec 2024

- Built a deep learning-based human pose classification system using TensorFlow, Keras, and MediaPipe to evaluate plank posture accuracy.
- Trained a multi-layer neural network with hyperparameter optimization via Keras Tuner, achieving reliable classification of correct, high-back, and low-back plank positions.
- Integrated real-time webcam detection with OpenCV and MediaPipe, and deployed the final model using a serialized .pkl pipeline for scalable inference.

Online Auction System

Jun.2025 – Jul. 2025

- Built a full-stack auction platform using React.js, Flask, and MongoDB, with support for item listings, bidding, real-time notifications, and payment tracking.

- Implemented JWT-based authentication, dynamic bidding logic with highest-bid validation, and a notification system for outbids, wins, and payment updates.

Student-Faculty Project Management System

Feb. 2025 – May 2025

- Developed a role-based academic project management platform using FastAPI and HTML/CSS/JS, enabling students to apply for and track faculty-guided research projects.
- Implemented CSV-based persistence for user authentication, project creation, application tracking, and meeting scheduling across student and faculty modules.
- Generated dynamic statistical reports and ensured session-managed login/logout flows with real-time data updates.

Snake Game

Aug. 2024 - Dec. 2024

- Designed and developed a classic Snake game using Java, implementing OOP(Object-Oriented Programming) principles.
- Utilized Java Swing and AWT to create an interactive GUI with smooth animations and responsive.

EXPERIENCE

Lipon | *Software Developer Intern*

June 2025 – July 2025

Worked as part of a 3-member team to develop a full-stack online auction platform using React.js, Flask, and MongoDB during a 45-day internship program.

Navatej Technologies | *AI Engineer Intern*

July 2025 – Present

I worked on advanced computer vision and data visualization tools, integrating LiDAR analysis, image measurement, and compliance automation. I worked on a Gradio-based labeling tool and a Streamlit analytics dashboard for real-time spatial measurements and 3D visualization. My work involved OpenCV, Open3D, Flask, and React to automate NESC compliance checks, annotation quality validation, and data governance workflows, combining AI-driven insights with robust software engineering practices.

EXTRA-CURRICULARS

Sakhi Club | *School of Engineering Representative*

Nov. 2024 – May 2025

Promote women empowerment and welfare initiatives through events, workshops, and mentorship programs.

CERTIFICATIONS

AI Tools and ChatGPT Workshop | *be10X*

Jan 2026

Practical training in prompt engineering, AI-assisted coding and debugging, rapid data analysis, workflow automation, and AI-driven content generation.

Artificial Intelligence and Machine Learning | *AcmeGrade*

June 2023-July 2023

I learned various libraries involved in python which help us work with different Machine Learning Models starting with the basic Numpy, Pandas to Sklearn library.

TCS iON Career Edge - Young Professional | *TCS iON*

Oct 2023 – Jan 2024

This course covered essential professional skills, including communication, presentation, soft skills, resume writing, and interview techniques. It also included business etiquette, email and telephone etiquette, accounting fundamentals, IT foundational skills, and an overview of artificial intelligence.

ACHIEVEMENTS

Academic Scholarship | *Mahindra University*

2023 – 2025

Secured 3rd Place at Aminoverse Hackathon | *Mahindra University*

Apr. 2025